

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 3**

**COURSE NAME: CSA0704 COURSE CODE: Computer Networks**

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### **COURSE OUTCOME COVERED**

**CO1:** *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

*Assessment Tool Weightage: 30%*

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Annexure A

### **TECHNICAL PRESENTATION**

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#### ***Code of Conduct:***

*I LAKSHITHA SREE K(192511336) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

## Annexure A

### TECHNICAL PRESENTATION

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

### RUBRICS FOR CALCULATION

| Criteria                  | Weight age | Level 4 – Exemplary                                                                           | Level 3 – Proficient                                        | Level 2 – Developing                                | Level 1 – Beginning                              |
|---------------------------|------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Understanding of Concepts | 25         | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design       | 30         | Well-planned, systematic implementation with optimal solution                                 | Correct implementation with minor issues                    | Basic implementation with errors or inefficiencies  | Incorrect or incomplete implementation           |
| Use of Tools              | 20         | Effective and advanced use of tools/technologies with proper configuration                    | Appropriate use of tools with correct execution             | Limited or inconsistent use of tools                | Incorrect or minimal use of tools                |

|                     |    |                                                                            |                                                |                                          |                                       |
|---------------------|----|----------------------------------------------------------------------------|------------------------------------------------|------------------------------------------|---------------------------------------|
| Analysis of Results | 15 | Insightful analysis with accurate interpretation and validation of results | Good analysis with reasonable interpretation   | Basic analysis with limited insights     | Poor or incorrect analysis of results |
| Documentation       | 10 | Clear, well-structured documentation with diagrams, code, and results      | Organized documentation with necessary details | Basic documentation with missing details | Poor or incomplete documentation      |

# ARTIFICIAL INTELLIGENCE FOR NETWORK TRAFFIC MANAGEMENT

PRESENTED BY

**LAKSHITHA SREE K**

REG NO: 192511336

# AI IN NETWORK TRAFFIC MANAGEMENT

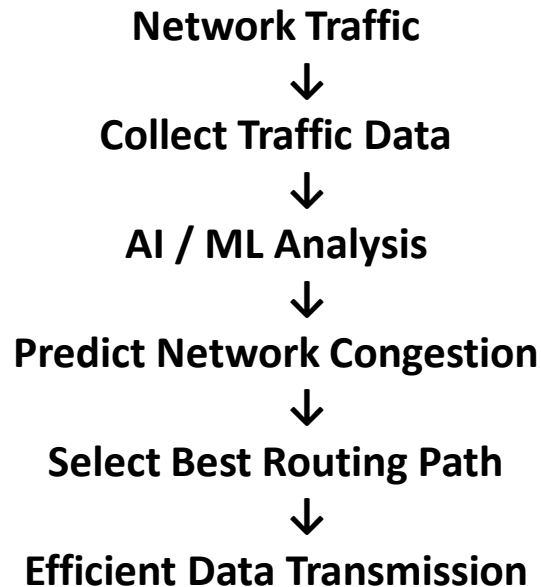
- AI monitors network traffic in real time.
- Predicts congestion before it occurs.
- Automatically selects the best routing path.
- Improves speed, reliability and bandwidth utilization.

## KEY COMPONENTS

- Traffic Monitoring
- Machine Learning
- Traffic Prediction
- Intelligent Routing
- Network Optimization

# WORKING PROCESS

## FLOWCHART



### Explanation:

- AI continuously collects traffic information, analyzes it using machine learning, predicts congestion, and reroutes packets through the best path to reduce delay and improve reliability.

# ADVANTAGES, APPLICATIONS & CONCLUSION

## **Advantages**

- Reduces congestion
- Better bandwidth utilization
- Low latency
- Automatic fault detection
- Improved QoS

## **Applications**

- Cloud Computing
- 5G Networks
- IoT
- Smart Cities
- Data Centers

## **Conclusion**

- Artificial Intelligence improves network efficiency by predicting congestion, optimizing routing decisions, reducing packet loss, and ensuring reliable communication. It is a key technology for future intelligent networks.